SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

# ENG201

**Timed Online Assignment – July Semester 2021**

## Linear Systems Analysis and Design

**Monday, 15 November 2021** **10:00 am – 12:30 pm**

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**Time allowed: 2.5 hours**

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**INSTRUCTIONS TO STUDENTS:**

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOACourse Code, Your Name and Student PI No. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT space. Example, TOAENG201_RaphaelLee_T1923161 (omit D/O, S/O).

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. All answers are to be typed in Times New Roman font, size 12 and single spacing. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. Flowcharts and graphs may be scanned or photographed and embedded in the Word file in the right sequence provided it does not exceed the file size limit of 500MB. All uploaded handwritten answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

7. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

8. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

*Answer all questions. (Total 100 marks)*

**Question 1**

(a) Determine whether the following signals are periodic. If a signal is periodic, then calculate its fundamental period.

(i) $x_1[n] = e^{j(\pi/4)n}$.

(2 marks)

(ii) $x_2[n] = \cos\left(\frac{1}{4}\right)n$.

(2 marks)

(b) A linear system, L, has the following relationship,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-2k],$$

between its input $x[n]$ and its output, $y[n]$, where $h[n] = u[n] - u[n-4]$ and $u[n]$ is the discrete unit step.

(i) Determine $y[n]$ when $x[n] = \delta[n-1]$.

(2 marks)

(ii) Determine $y[n]$ when $x[n] = \delta[n-2]$.

(2 marks)

(iii) Is the system linear time-invariant (LTI)?

(4 marks)

(iv) Determine and sketch $y[n]$ when $x[n] = u[n]$.

(5 marks)

(c) Which of the following impulse responses corresponds to a stable LTI system? Justify your answer.

(i) $h_1[n] = n\cos\left(\frac{\pi n}{4}\right)u[n]$.

(4 marks)

(ii) $h_2[n] = 3^n u[-n+10]$.

(4 marks)

**Question 2**

Consider the following three linear time-invariant (LTI) systems connected as shown in Figure Q2 below:

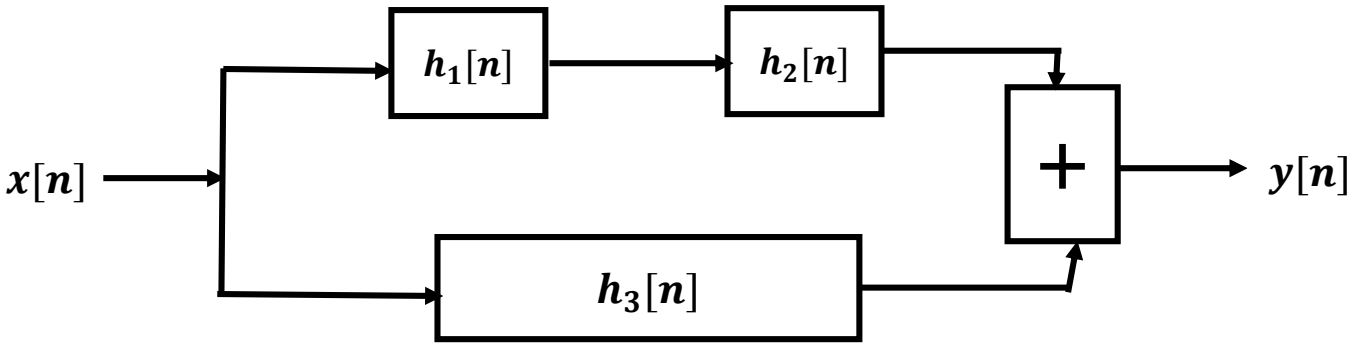


**Figure Q2**

(a) The impulse response of each block is given by:

$$h_1[n] = -0.8\delta[n] + \delta[n-1] + \delta[n-2] - 0.8\delta[n-3],$$

$$h_2[n] = 0.5\delta[n] + 0.5\delta[n-1] + 0.5\delta[n-2],$$

and

$$h_3[n] = 0.1\delta n] - 0.1\delta[n-2] - 0.1\delta[n-3] + 0.1\delta[n-4] + 0.2\delta[n-5].$$

Describe the overall system impulse response $h_s[n]$.

(10 marks)

(b) Analyse the above system's frequency response by computing the Discrete Fourier Transform of its impulse response.

(10 marks)

(c) An input signal $x[n] = 2\cos\left(\frac{2\pi}{6}n\right)$ is fed into the above system. From the frequency response obtained in Question 2(b), calculate the output signal, $y[n]$.

(3 marks)

(d) Discuss the stability of the above given system.

(2 marks)

**Question 3**

(a) Given a linear time-invariant (LTI) system described by the differential equation

$$\frac{d^2y(t)}{dt^2} - 2\frac{dy(t)}{dt} + y(t) = 2\cos t, \text{ where } y(0) = 1, \frac{d}{dt}y\left({}^{3\pi}/_{2}\right) = 2.$$

(i) Solve and obtain the homogeneous solution for the above LTI system.

(4 marks)

(ii) Compute the particular solution of the above system.

(5 marks)

(iii) Design the overall solution to the above system.

(6 marks)

(b) Consider a LTI system described by the following first-order initial value problem:

$$\frac{dy(t)}{dt} + 5y(t) = g(t), y(0) = 1,$$

where

$$g(t) = \begin{cases} 0, & if\ 0 \le t < 1, \\ 5t, & if\ t \ge 1. \end{cases}$$

Use Laplace Transform to find the unique solution $y(t)$ for the above problem.

(10 marks)

**Question 4**

(a) Prove from the first principles of z-transform that:

$$\{x[k+m]\} \leftrightarrow z^m[X(z) - I(z)],$$

where

$$X(z) = \sum_{k=0}^{\infty} x(k)z^{-k},$$

and

$$I(z) = \sum_{k=0}^{m-1} x(k)z^{-k}.$$

(10 marks)

(b) Using the relationship obtained in Question 4(a), determine the z-transform of the sequence $\{y[n]\}$ if

$$y[n+3] + 2y[n+2] + 3y[n+1] + y[n] = u[n-1],$$

where $u[n]$ is a unit-step function and $y[0] = 1, y[1] = -1, y[2] = -3$.

(15 marks)

**----- END OF PAPER -----**